



GSoC'26 Final Project Report

A Reference-Free “Generate Once, Synthesize Many”
Framework for Synthetic Anomaly Generation in
Industrial Anomaly Detection.

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ORGANIZATION

OpenVINO Toolkit | Intel Benelux B.V.

CONTRIBUTOR

NAME: Abhay Kumar Das

PHONE: +91 8340574215

EMAIL: dasabhay.jsr@gmail.com



FLASH: A Reference-Free “Generate Once, Synthesize Many” Framework for Synthetic Anomaly Generation in Industrial Anomaly Detection

Problem Statement

Industrial anomaly detection systems require defective samples for effective training, evaluation, and threshold calibration. However, real-world defects are often **rare, diverse, expensive to collect, and difficult to annotate**. This becomes especially challenging in zero-defect-reference settings where no real anomalous image or defect mask is available.

Existing synthetic anomaly generation methods generally fall into two categories. **Procedural methods** are fast and computationally efficient but are often limited to predefined noise, texture, or structural perturbations. **Generative methods** can produce more diverse and semantically meaningful defects, but generating a new defect for every image is computationally expensive.

The key challenge is therefore:

“How can realistic and diverse anomalies be generated from only normal images while avoiding repeated expensive generative inference?”

Solution

FLASH (Generate Once, Synthesize Many) is a reference-free synthetic anomaly generation framework designed to separate **defect generation** from **anomaly synthesis**.

Instead of generating a new anomaly independently for every image, FLASH generates a limited number of semantic defects once, extracts and validates them, and stores them in a **category-wise Semantic Defect Bank**. These reusable defects can then be placed on different normal images at different locations and spatial scales.

This allows FLASH to combine the **semantic richness of generative models** with the **efficiency and scalability of procedural synthesis**, without requiring real defect references.

Methodology

1. Semantic-Guided Anomaly Generation

A normal image is analyzed by **VLM-1 (Qwen2.5-VL-7B-Instruct)** to determine a plausible category-specific anomaly. The generated anomaly description is combined with a configurable dataset context and hard generation constraints before being provided to the image-generation model.

The output is a **Generated Anomaly Image** containing the synthesized defect.

2. Defect Extraction

The generated anomaly cannot directly be used as a reusable defect because the generation model may introduce small spatial, geometric, or photometric differences.

FLASH therefore uses **DiffMask** to compare the normal image and generated anomaly. Registration, photometric adjustment, multiple difference cues, local statistics, adaptive thresholding, and refinement are used to recover the **actual generated defect mask**.

3. Defect Validation & Storage

The extracted defect is passed to **VLM-2** for semantic validation. The system verifies whether the detected region represents a plausible category-specific anomaly rather than reconstruction noise or an irrelevant image difference.

Validated defects are stored in a **Category-wise Semantic Defect Bank**, allowing the same generated defect to be reused across multiple normal images.

4. Object-Aware Localization

For a new normal host image, FLASH first determines where a defect can physically occur.

Object Boundary Suppression (OBS) estimates the foreground object region using colour and texture information, preventing defects from being placed on the background.

Within this object region, **Multi-Resolution Spectral Pyramid (MRSP)** generates coherent, size-controllable placement masks. MRSP provides variation in defect location and spatial extent without requiring another generative-model call.

5. Adaptive Synthesis

A validated defect is retrieved from the Semantic Defect Bank and placed onto a different normal host image according to the MRSP placement region.

FLASH adapts the defect's **scale, orientation, position, and containment** to the selected region and harmonizes it with the host image. Finally, **hybrid blending** using feathered alpha blending or Poisson blending produces the final synthetic anomaly while preserving the surrounding image.

Architecture

Place the provided 5-stage FLASH architecture diagram here, immediately after the Methodology introduction and before the five stage explanations.

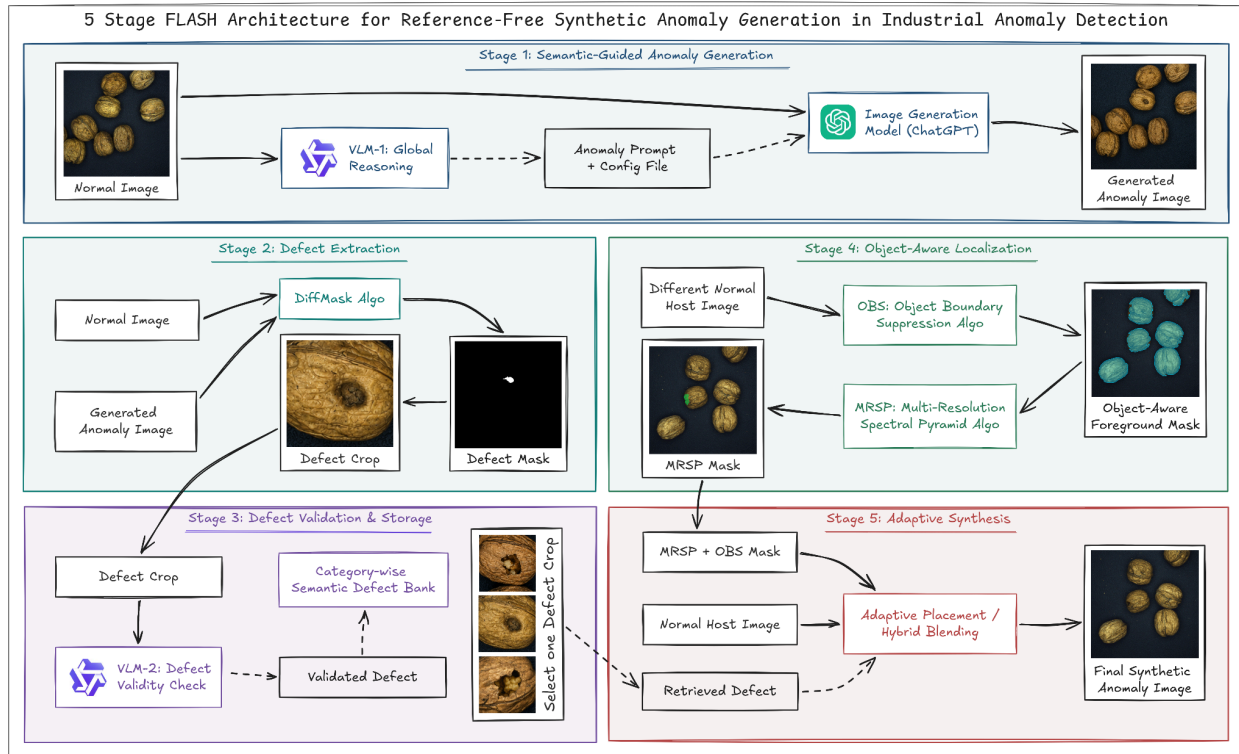


Figure: Five-Stage FLASH Architecture for Reference-Free Synthetic Anomaly Generation.

The diagram should be the **main architecture figure of the report**, because it gives the reviewer the complete pipeline before the individual stages are explained. Your current architecture clearly shows the separation between **Stages 1–3: defect generation and banking** and **Stages 4–5: reusable defect synthesis**.

Project Deliverables

GitHub Repository:

<https://github.com/AbhayKumarDas/flash>

Project Documentation / Notion:

[GSoC'26](#) | [Weekly Updates](#) | [Anomalib](#) | [Notion](#)

Final Presentation:

 GSoC'26 Project

Final Paper Submission:

Updated Soon...

Conclusion

FLASH demonstrates a practical approach to reference-free synthetic anomaly generation by separating **what defect is generated** from **where and how that defect is synthesized**. By generating and validating defects once and subsequently reusing them across different normal host images, the framework reduces repeated generative inference while maintaining controllable anomaly diversity.

The project establishes a foundation for scalable synthetic anomaly generation and demonstrates that generated anomalies can provide useful calibration signals even when real defect samples are unavailable. Future work focuses on improving category-dependent calibration gaps, increasing defect-bank diversity, and extending evaluation to additional industrial domains.